Range of (x-2)/(x-3/

Let's carefully find the **range** of

$$f(x) = \frac{x-2}{x-3}$$

### Step 1: Identify domain

Denominator cannot be zero:

$$x \neq 3$$

So the function is defined for all real $x$ except $x = 3$.

### Step 2: Express in another form

Divide numerator and denominator:

$$f(x) = \frac{x-3+1}{x-3} = 1 + \frac{1}{x-3}$$

### Step 3: Analyze the expression

So,

$$f(x) = 1 + \frac{1}{x-3}$$

- Here, $\frac{1}{x-3}$ can take **any real value except 0**.
- That means $f(x)$ can take **any real value except 1**.

### Step 4: Confirm

If we try $f(x) = 1$:

$$1 = \frac{x-2}{x-3}$$

Cross-multiply:

$$x - 2 = x - 3 \quad \Longrightarrow \; -2 = -3$$

❌ Contradiction. So $y = 1$ is impossible.

### ✅ Final Answer:

The range of $\dfrac{x-2}{x-3}$ is

$$\mathbb{R} \setminus \{1\}$$

i.e., all real numbers except $1$.

Do you want me to also show the **graph approach** (asymptotes and curve sketching) so it's visually clear why $1$ is excluded?

How we know that range can't have 1 and also give me the technique to find domain and range of any kind of function

Good question 👍 Let's go step by step.

## 1. Why the range of $\frac{x-2}{x-3}$ cannot have 1

We want to see if the function can ever output $1$.

Set

$$y = \frac{x-2}{x-3}$$

If the function could be $1$, then

$$1 = \frac{x-2}{x-3}$$

Cross multiply:

$$x - 2 = x - 3$$

This simplifies to

$$-2 = -3$$

❌ Contradiction.
That means **there is no value of $x$ that makes the output equal to 1**.
So $y = 1$ is excluded from the range.

Another way:
Rewrite

$$f(x) = 1 + \frac{1}{x-3}$$

Here $\frac{1}{x-3}$ is **never equal to 0** (since denominator can't be infinite).
So $f(x)$ is never exactly $1$.

That's how we *know* range doesn't include $1$.

## 2. General Technique to Find Domain

To find the domain (allowed $x$ values):

- **Polynomials** (e.g., $x^2 + 3x + 1$): domain = all real numbers.
- **Rational functions** (fractions): exclude values that make denominator = 0.
- **Square roots/even roots**: inside must be $\geq 0$ (for real functions).
- **Logarithms**: argument must be $> 0$.
- **Trigonometric functions**: exclude points where undefined (e.g., $\tan x$ excludes $\frac{\pi}{2} + n\pi$).

👉 Rule of thumb: **Look for restrictions** (denominator = 0, negative under even root, log of non-positive, etc.).

## 3. General Technique to Find Range

Ways to find range:

1. **Solve for $x$ in terms of $y$**:
   Take $y = f(x)$, then solve for $x$. Check for values of $y$ that lead to contradictions → those values are excluded from the range.
   (We did this with $y = (x-2)/(x-3)$).
2. **Transform/Rewrite the function**:
   Example: $\frac{x-2}{x-3} = 1 + \frac{1}{x-3}$. This helps see which values are possible or not.
3. **Graphical method**:
   Sketch the function (asymptotes, intercepts). Look at how the curve behaves → the missing horizontal/vertical lines tell you about excluded values in the range.
4. **Monotonic behavior**:
   If a function is continuous and monotonic on intervals, you can find range by checking the limits at boundaries and at infinity.

### ✅ Summary

- To find **domain**: check input restrictions (denominator, square root, log, trig undefined points).
- To find **range**: either
  - invert $y = f(x)$ and check which $y$ are possible, or
  - rewrite function to see excluded values, or
  - use graphing/limits.

Do you want me to make a **step-by-step recipe (flowchart style)** for finding domain & range of *any* given function?

Same kind of other examples of rational fraction

Perfect 👍 Let's practice with a few more **rational functions (fractions)** step by step, just like we did for $\frac{x-2}{x-3}$.

### ◆ Example 1:

$$f(x) = \frac{x+1}{x-1}$$

**Domain:** Denominator $x - 1 = 0 \implies x \neq 1$.
**Range:** Let

$$y = \frac{x+1}{x-1}$$

Cross multiply:

$$\begin{aligned} y(x-1) = x + 1 \quad &\Longrightarrow \quad yx - y = x + 1 \\ yx - x = y + 1 \quad &\Longrightarrow \quad x(y-1) = y + 1 \\ x = \frac{y+1}{y-1}&, \quad y \neq 1 \end{aligned}$$

✅ So range = $\mathbb{R} \setminus \{1\}$.

### ◆ Example 2:

$$f(x) = \frac{2x}{x+3}$$

**Domain:** $x \neq -3$.
**Range:**
Let

$$y = \frac{2x}{x+3}$$

Cross multiply:

$$\begin{aligned} y(x+3) = 2x \quad &\Longrightarrow \quad yx + 3y = 2x \\ yx - 2x = -3y \quad &\Longrightarrow \quad x(y-2) = -3y \\ x = \frac{-3y}{y-2}&, \quad y \neq 2 \end{aligned}$$

✅ Range = $\mathbb{R} \setminus \{2\}$.

### ◆ Example 3:

$$f(x) = \frac{x}{x^2+1}$$

**Domain:** All real $x$ (denominator never zero).
**Range:**
Let

$$y = \frac{x}{x^2+1}$$

Cross multiply:

$$\begin{aligned} y(x^2+1) = x \quad &\Longrightarrow \quad yx^2 + y = x \\ yx^2 - x + y &= 0 \end{aligned}$$

This is a quadratic in $x$. For real $x$, discriminant $\Delta \geq 0$:

$$\begin{gathered} \Delta = (-1)^2 - 4(y)(y) = 1 - 4y^2 \geq 0 \\ |y| \leq \tfrac{1}{2} \end{gathered}$$

✅ Range = $[-1/2,\ 1/2]$.

### ◆ Example 4:

$$f(x) = \frac{1}{x}$$

**Domain:** $x \neq 0$.
**Range:** Solve

$$y = \frac{1}{x} \quad \Longrightarrow \quad x = \frac{1}{y}, \quad y \neq 0$$

✅ Range = $\mathbb{R} \setminus \{0\}$.

### ✨ General Pattern for Rational Functions

- If numerator and denominator are both **linear**, the range will usually be all reals **except one value**.
- If denominator is quadratic, range often comes out as an **interval** (like Example 3).
- Best trick: **Set $y = f(x)$, cross-multiply, solve for $x$, then see which $y$ values are excluded.**

👉 Do you want me to prepare **5 practice problems** for you (with hints but not solutions) so you can test this method yourself?